UNICEF

Using the UNICEF SDMX Data Webpage

Introduction

UNICEF provides free, public access to international statistics on children and women, including health, nutrition, mortality, immunisation, education and protection indicators.

The UNICEF SDMX data webpage allows you to explore these indicators and download the underlying data using a structured web system. You do not need to create an account, request permission or install software to get started.

If you can open a web page and copy and paste a link, you can use this system.

This guide explains how to:

- understand what the UNICEF SDMX webpage is showing
- find datasets and indicators
- identify the correct indicator codes
- retrieve data for specific countries and years
- avoid common mistakes

What this webpage is (in plain terms)

The UNICEF SDMX webpage is **not a normal data table**. It is a structured interface that sits on top of a database.

It shows you:

- what datasets exist
- how indicators are organised
- what codes must be used to request data

When you build a correct web address using these codes, UNICEF sends back a table of data in a structured format called JSON. Many tools including browsers and Excel can read this format.

The UNICEF SDMX data webpage address

This is the main page you will use:

https://sdmx.data.unicef.org/webservice/data.html

Paste this entire address into your browser and press Enter.

This page helps you **explore and build data requests**. It does not automatically show indicator values.

Understanding what you see on the page

The page is divided into sections, usually shown on the left side.

Key sections include:

- Dataflows
- Data Structures
- Codelists
- Data

These sections are how UNICEF organises its indicators and datasets.

What is a dataflow

A **dataflow** is a dataset. It groups together indicators that share the same structure.

For example, a dataflow might include:

- multiple child mortality indicators
- multiple nutrition indicators
- multiple immunisation indicators

Each dataflow has:

- an ID
- a name
- a version

You must know the dataflow ID before you can retrieve data.

Finding available dataflows

On the webpage:

1. Click **Dataflows**
2. Scroll through the list

3. Look at the Name column to find a relevant topic
4. Note the ID of the dataflow you want to use

This step is essential. Without a dataflow ID, you cannot download data.

Understanding indicator and dimension codes

Within each dataflow, data are organised using **dimensions**. These are categories such as:

- country
- indicator
- year
- sex
- age group

Each dimension uses **codes**, not full names.

Examples:

- Country may be coded as AUS instead of Australia
- Indicators may use short codes rather than full titles
- Year is often listed as TIME_PERIOD

To see valid codes:

1. Click **Data Structure** or **Codelists**
2. Find the dimension you need
3. Write down the exact codes shown

The codes must be used exactly as written.

How indicator identification works

Unlike the WHO GHO system, UNICEF indicators are usually selected **inside a dataflow**, not from one global indicator list.

This means:

- you first choose the dataflow
- then you choose the indicator code within that dataflow
- then you choose country and year

Indicator names are descriptive, but indicator codes are what actually retrieve the data.

Using the webpage to help build the request

The UNICEF SDMX webpage allows you to:

- click through dataflows
- inspect structures
- see dimension order
- test URLs before using them elsewhere

This reduces errors and makes it easier to build valid requests.

Filtering the data

By default, a request may return a large amount of data. You can narrow results by choosing specific dimension codes.

Common filters include:

- country
- indicator
- year

You must use the correct codes from the codelists.

How to use this with the GBD Pipeline

By using ctrl + F (windows) or command + F (mac), within the GBD_final code, the following phrase “XXXXXIndicator” will be just above the section to change.

```
# Define all indicators to be fetched - HERE YOU CAN CHANGE INDICATOR CODE AND TITLE FOR WHICHEVER DATASET YOU REQUIRE FOR THE UNDE
# Use the following structure XXXXXIndicator('INDICATOR_CODE', 'TITLE FOR INDICATOR ')

indicators = [
    # --- Malnutrition --- ALTER INDICATORS HERE
    WorldBankIndicator('SH.STA.MALN.ZS', 'Malnutrition (wght-for-age z<-2)'),
    WHOIndicator('NUTRITION_WH_2', 'Wasting prevalence among children under 5 years'),
    # Removed DHS indicators due to lack of available data in the DHS API as of July 2025.

    # --- Low Birth Weight --- ALTER INDICATORS HERE
    WorldBankIndicator('SH.STA.BRTW.ZS', 'Low birth weight (=<2500 g)'),
    # Removed DHS indicators due to lack of available data in the DHS API as of July 2025.

    # --- Breastfeeding --- ALTER INDICATORS HERE
    WHOIndicator('WHOSIS_000006', 'Exclusive breastfeeding under 6 months (%)'),

    # --- Household Environment --- ALTER INDICATORS HERE
    WorldBankIndicator('EG.USE.COMM.CL.ZS', 'Use solid fuels (yes)'),
    # Removed DHS indicators due to lack of available data in the DHS API as of July 2025.

    # --- Child Mortality (for stratification) --- ALTER INDICATORS HERE
    WorldBankIndicator('SH.DYN.MORT', 'Child Mortality Rate (U5MR)'),

    # --- Crowding & Housing --- ALTER INDICATORS HERE
    UNHabitatIndicator(),
    WorldBankIndicator('EN.POP.SLUM.UR.ZS', 'Urban slum population (% of urban)'),

    # --- Population Data (absolute counts 0-4) --- POPULATION DO NOT ALTER
    UNICEFUnder5PopIndicator(),
]
```

Here any of the blue can be changed with the left ‘ ‘ refers to the indicator code and the right ‘ ‘ refers to the title (title can be named anything).

Common problems and fixes

Problem: No data returned
Reason: One or more codes are invalid
Fix: Check the codelists and spelling exactly

Problem: Error message in browser
Reason: Incorrect dataflow ID or version
Fix: Recheck the dataflow information

Problem: Too much data
Fix: Add country or year filters

Problem: Cannot find an indicator
Fix: Check a different dataflow. Indicators are grouped by topic.

WHO

Using the WHO Global Health Observatory (GHO) API

Introduction

The World Health Organization Global Health Observatory (GHO) provides free, public access to international health statistics. These data include mortality, disease burden, risk factors, service coverage and many other indicators used in clinical practice, research and policy.

The GHO Application Programming Interface (API) allows you to access these data directly through a web address. You do not need to create an account, request permission or install software to get started.

If you can open a web page and copy and paste a link, you can use this API.

This guide explains how to:

- find available WHO indicators
- identify the correct indicator code
- retrieve data for that indicator
- narrow the data by country, year or sex

What is an API (in plain terms)

An API is simply a structured web address that returns data instead of a normal web page.

When you paste a GHO API address into your browser, WHO sends back a table of data in a structured text format called JSON. Many tools including browsers, Excel, R and Python can read this format. This format will require the pretty print to checked and you are looking for the indicator ID.

You do not need to understand JSON to use the data.

The GHO Indicator API base address

This is the starting point for finding all available indicators:

https://ghoapi.azureedge.net/api/Indicator

Paste this entire address into your browser and press Enter.

You will see a long list of entries. Each entry represents one WHO indicator.


```
    "Language": "EN"
  },
  {
    "IndicatorCode": "GDO_q13x3",
    "IndicatorName": "Existence of at least one functioning risk reduction campaign",
    "Language": "EN"
  },
  {
    "IndicatorCode": "GOE_Q203",
    "IndicatorName": "Lack of integration between various health services",
    "Language": "EN"
  },
  {
    "IndicatorCode": "HCF_REL_ELECTRICITY",
    "IndicatorName": "Percentage of health-care facilities with reliable electricity supply (%)",
    "Language": "EN"
  },
  {
    "IndicatorCode": "HIV_0000000007",
    "IndicatorName": "Testing and counselling facilities, reported number",
    "Language": "EN"
  },
  {
    "IndicatorCode": "HIV_0000000008",
    "IndicatorName": "Testing and counselling facilities, estimated number per 100 000 adult population",
    "Language": "EN"
  },
  {
    "IndicatorCode": "HIV_0000000015",
    "IndicatorName": "Number of pregnant women living with HIV who received antiretrovirals for preventing mother-to-child transmission",
    "Language": "EN"
  },
  {
    "IndicatorCode": "HIV_0000000016",
    "IndicatorName": "Number of pregnant women living with HIV who received antiretrovirals for preventing mother-to-child transmission",
    "Language": "EN"
  },
  {
    "IndicatorCode": "HIV_0000000022",
    "IndicatorName": "Estimated number of people eligible for antiretroviral therapy according to 2010 guidelines",
    "Language": "EN"
  },
  {
    "IndicatorCode": "MENING_1",
    "IndicatorName": "Number of suspected meningitis deaths reported",
    "Language": "EN"
  },
  {
    "IndicatorCode": "MENING_2",
    "IndicatorName": "Number of suspected meningitis cases reported",
    "Language": "EN"
  },
  {
    "IndicatorCode": "MH_10",
    "IndicatorName": "Mental health outpatient facilities (per 100,000)",
    "Language": "EN"
  },
  {
    "IndicatorCode": "MH_11",
```

Understanding what you see

Each indicator entry includes:

- IndicatorCode
  This is the unique identifier you must use to download data.
- IndicatorName
  This is the human-readable name of the indicator.
- Language

Example indicator names include:

- Life expectancy at birth

- Maternal mortality ratio
- Ambient air pollution attributable deaths

The IndicatorCode is essential. Without it, you cannot retrieve the data.

How to use this with the GBD Pipeline

By using ctrl + F (windows) or command + F (mac), within the GBD_final code, the following phrase “XXXXXIndicator” will be just above the section to change.

```python

# Define all indicators to be fetched - HERE YOU CAN CHANGE INDICATOR CODE AND TITLE FOR WHICHEVER DATASET YOU REQUIRE FOR
# Use the following structure XXXXXIndicator('INDICATOR_CODE', 'TITLE FOR INDICATOR ')

indicators = [
    # --- Malnutrition --- ALTER INDICATORS HERE
    WorldBankIndicator('SH.STA.MALN.ZS', 'Malnutrition (wght-for-age z<-2)'),
    WHOIndicator('NUTRITION_WH_2', 'Wasting prevalence among children under 5 years'),
    # Removed DHS indicators due to lack of available data in the DHS API as of July 2025.

    # --- Low Birth Weight --- ALTER INDICATORS HERE
    WorldBankIndicator('SH.STA.BRTW.ZS', 'Low birth weight (=<2500 g)'),
    # Removed DHS indicators due to lack of available data in the DHS API as of July 2025.

    # --- Breastfeeding --- ALTER INDICATORS HERE
    WHOIndicator('WHOSIS_000006', 'Exclusive breastfeeding under 6 months (%)'),

    # --- Household Environment --- ALTER INDICATORS HERE
    WorldBankIndicator('EG.USE.COMM.CL.ZS', 'Use solid fuels (yes)'),
    # Removed DHS indicators due to lack of available data in the DHS API as of July 2025.

    # --- Child Mortality (for stratification) --- ALTER INDICATORS HERE
    WorldBankIndicator('SH.DYN.MORT', 'Child Mortality Rate (U5MR)'),

    # --- Crowding & Housing --- ALTER INDICATORS HERE
    UNHabitatIndicator(),
    WorldBankIndicator('EN.POP.SLUM.UR.ZS', 'Urban slum population (% of urban)'),

    # --- Population Data (absolute counts 0-4) --- POPULATION DO NOT ALTER
    UNICEFUnder5PopIndicator(),
]
```

Here any of the blue can be changed with the left ‘ ‘ refers to the indicator code and the right ‘ ‘ refers to the title (title can be named anything).

Finding the indicator you want

Because there are many indicators, it is best to filter the list.

Search by keyword in the indicator name

To search for indicators containing a specific word, add a filter to the address.

Example: find indicators containing the word mortality

https://ghoapi.azureedge.net/api/Indicator?$filter=contains(IndicatorName,'mortality')

Example: find indicators containing diabetes

https://ghoapi.azureedge.net/api/Indicator?$filter=contains(IndicatorName,'diabetes')

Scroll through the results and note the IndicatorCode for the indicator you want.

Filtering the data

By default, the API returns all data. You can narrow results using filters.

Filter by sex

For many indicators:

- MLE = Male
- FMLE = Female
- BTSX = Both sexes

Example: male only life expectancy

https://ghoapi.azureedge.net/api/WHOSIS_000001?$filter=Dim1 eq 'MLE'

Filter by country

Countries are usually identified by ISO codes.

Example: Australia

https://ghoapi.azureedge.net/api/WHOSIS_000001?$filter=SpatialDim eq 'AUS'

If a filter returns no data, that indicator may not support that dimension.

Filter by year

Years are based on the time dimension start date.

Example: data for year 2019

https://ghoapi.azureedge.net/api/WHOSIS_000001?$filter=year(TimeDimensionBegin) eq 2019

Combine filters

You can combine filters using the word and.

Example: Australia, males, 2019

https://ghoapi.azureedge.net/api/WHOSIS_000001?$filter=SpatialDim eq 'AUS' and Dim1 eq 'MLE' and year(TimeDimensionBegin) eq 2019

Common problems and fixes

Problem: You see no data
Reason: The indicator does not support that filter
Fix: Remove one filter and try again

Problem: Error message in browser
Reason: Typo in the address
Fix: Check spelling and quotation marks

Problem: Too much data
Fix: Add filters for country, year or sex

WORLDBANK

Using the World Bank Indicator Webpage

Introduction

The World Bank provides free, public access to international development and health statistics covering mortality, disease burden, health systems, population, economics and social determinants of health.

The World Bank indicator webpage allows you to **browse, search and identify indicators** using a simple web interface. Each indicator has a unique code that can be used for analysis, reporting or downstream data pipelines.

You do not need to create an account, request permission or install software to use this webpage.

If you can open a web page, scroll and copy text, you can use this resource.

This guide explains how to:

- navigate the World Bank indicator webpage
- find indicators by topic or keyword
- identify the correct indicator code
- understand what the indicator code represents
- avoid common mistakes when selecting indicators

What this webpage is (in plain terms)

The World Bank indicator webpage is a **catalogue of indicators**.

Each page represents:

- one indicator
- one definition
- one unique indicator code

The webpage is designed to be human readable and is easier to use than most statistical systems. You do not need to understand programming or data formats to find what you need.

The indicator code is the most important output from this page.

The World Bank indicator webpage address

This is the main page you will use:

https://data.worldbank.org/indicator

Paste this entire address into your browser and press Enter.

You will see a searchable list of indicators organised by topic.

Understanding what you see on the page

The page includes:

- a search bar at the top
- topic categories such as Health, Population, Economy and Environment
- a scrolling list of indicator names

Each indicator name is clickable and opens a dedicated indicator page.

What an indicator code is

Each World Bank indicator has a **unique indicator code**.

The code:

- is short and structured
- does not change over time
- uniquely identifies that indicator across all countries and years

Examples include:

- SP.DYN.LE00.IN for life expectancy at birth
- SH.DYN.MORT for under five mortality rate
- SH.IMM.MEAS for measles immunisation coverage

The indicator code is essential. Without it, you cannot reliably identify or reuse the indicator.

Finding an indicator using search

To find an indicator:

1. Go to the indicator webpage
2. Click in the search bar
3. Type a keyword such as mortality, diabetes, vaccination or life expectancy
4. Press Enter or wait for results to appear
5. Click the indicator name that best matches your needs

This is the fastest way to locate a specific indicator.

Finding indicators by topic

You can also browse by topic:

1. Scroll down the main page
2. Select a topic such as Health, Population or Nutrition
3. Review the list of indicators under that topic
4. Click the indicator name to open its page

This is useful when you are exploring rather than searching for a specific variable.

Understanding an indicator page

When you click an indicator, you will see:

- the indicator name
- a short definition
- the indicator code
- units of measurement
- source information

The **indicator code is usually displayed near the top** of the page. This is the code you should copy.

How to use this with the GBD Pipeline

By using ctrl + F (Windows) or command + F (Mac) within the GBD_final code, the following phrase “XXXXXIndicator” will appear just above the section to change.

Here any of the blue text can be changed. The left single quotation marks refer to the **indicator code** and the right single quotation marks refer to the **indicator title**. The title can be named anything and does not affect data extraction.

Only the indicator code must be exact.

Choosing the correct indicator

Many indicators sound similar but are not identical.

Before selecting an indicator:

- read the definition carefully
- check the units
- check whether it is a rate, proportion or absolute count
- confirm the population group

For example, mortality indicators may differ by age group, cause or denominator.

Common problems and fixes

Problem: Multiple indicators appear to measure the same thing
Reason: Different definitions or populations
Fix: Read the indicator description carefully and choose the most appropriate one

Problem: Indicator name looks right but results differ from expectations
Reason: Units or age group are different
Fix: Check units and population coverage on the indicator page

Problem: Cannot find a clinical indicator
Reason: The World Bank focuses on population level measures
Fix: Try broader public health or health system indicators

DHS

Using the DHS Program Indicators Webpage

Introduction

The Demographic and Health Surveys (DHS) Program provides publicly accessible health and demographic indicators collected from nationally representative household surveys in many low and middle income countries.

The DHS indicators webpage lists **all available DHS indicators**, along with their unique indicator codes and definitions. These indicator codes can then be used in analysis pipelines, scripts or structured workflows such as Global Burden of Disease (GBD) aligned analyses.

You do not need programming skills to use this webpage. If you can open a browser, scroll and copy text, you can use it.

This guide explains how to:

- understand what the DHS indicators webpage shows
- find and interpret DHS indicator codes
- select the correct indicator using definitions
- use DHS indicator codes in GBD style analysis pipelines
- avoid common mistakes

What this webpage is (in plain terms)

This webpage is a **catalogue of DHS indicators**.

Each row represents one indicator and shows:

- the indicator code
- a short label
- a formal definition

The indicator code is the most important element. It is the unique identifier used when extracting DHS data programmatically or within automated analysis workflows.

You do not need to understand APIs or data formats to identify indicator codes on this page.

The DHS indicators webpage address

This is the page you will use:

https://api.dhsprogram.com/rest/dhs/indicators?returnFields=IndicatorId,Label,Definition&f=html

Paste this entire address into your browser and press Enter.

A long table will load containing all DHS indicators.

Understanding what you see on the page

Each row contains three columns:

IndicatorId
This is the **unique indicator code**. Example: FE_FRTR_W_TFR

Label
This is the human readable indicator name. Example: Total fertility rate 15–49

Definition
This explains exactly what the indicator measures, including age group, denominator and units.

The IndicatorId must be copied exactly as shown.

Finding the indicator you want

Because the list is long, use your browser search.

Steps:

1. Press Ctrl + F (Windows) or Command + F (Mac)
2. Type a keyword such as mortality, stunting, malaria, contraceptive or anaemia
3. Scroll through matches and read the labels and definitions
4. Identify the indicator that best matches your analytic need
5. Copy the IndicatorId

Understanding indicator codes

DHS indicator codes follow a consistent pattern:

- Prefixes indicate domain
  FE = fertility
  CH = child health
  ML = malaria
  NT = nutrition
- Middle sections often indicate population
  W = women
  C = children
  M = men
- Suffixes indicate the specific measure

Codes are case sensitive and must not be altered.

Why indicator definitions matter

Many DHS indicators sound similar but differ in:

- age range
- denominator
- survey population
- whether the measure is a rate, proportion or mean

Always read the definition column before selecting an indicator code. This step prevents misalignment in downstream analyses.

How to use this with the GBD analysis pipeline

In GBD aligned or GBD style analysis pipelines, DHS indicators are usually referenced by their **IndicatorId**.

To update or change a DHS indicator in the pipeline:

1. Open the GBD_final script or configuration file
2. Use Ctrl + F (Windows) or Command + F (Mac)
3. Search for the placeholder text
   XXXXXIndicator

This placeholder appears immediately above the section where DHS indicators are defined.

Editing the indicator section

In the indicator definition section, you will typically see a structure similar to:

'FE_FRTR_W_TFR' = 'Total fertility rate'

In this structure:

- the text inside the **left single quotation marks** is the DHS IndicatorId
- the text inside the **right single quotation marks** is a human readable title

Only the indicator code must be exact.
The title can be changed freely and does not affect data extraction.

Updating the indicator

To update the indicator:

1. Replace the existing IndicatorId with the new one copied from the DHS webpage
2. Optionally update the title to match the new indicator
3. Save the file and rerun the pipeline

No other changes are required if the indicator code is valid.

Common problems and fixes

Problem: No data returned in analysis
Reason: IndicatorId is incorrect or misspelled
Fix: Copy and paste the IndicatorId directly from the DHS webpage

Problem: Indicator runs but results are unexpected
Reason: Wrong age group or denominator
Fix: Recheck the indicator definition

Problem: Too many similar indicators
Reason: DHS often reports the same concept across populations
Fix: Choose the indicator that matches the target population used in GBD